

Ex. 1 + Rés. : Soit

$$f: \mathbb{Z}_2 \oplus \mathbb{Z}_2 \rightarrow \mathbb{N}_5,$$

$f \rightarrow$ morph. de l'alg.
morph. $\Leftrightarrow$

$$\Leftrightarrow \begin{aligned} & (x+y) \in \{0, 1, 2, 3\} \\ & f(x+y) = f(x) \vee f(y) \\ & f(x \wedge y) = f(x) \wedge f(y) \\ & f(0) = 0, f(1) = 1 \end{aligned}$$

$$\Rightarrow (X, \gamma \in \{0, 1\}^n) \\ (X \leq Y \Rightarrow f(X) \leq f(Y))$$

$$f(a \vee b) = f(a) \vee f(b)$$

$$f(a \wedge b) = f(a) \wedge f(b)$$

$$f(0) = 0 \quad f(1) = 1$$

$\Rightarrow$ f este un homom f este un homom f este un homom

3

